

Multiple Choice (10 marks)

1. Are Python variable names case-sensitive?
 - (a) Yes
 - (b) No
 - (c) It's machine-dependent
 - (d) None of the above
2. Which of the following is correct about Python?
 - (a) It supports automatic garbage collection.
 - (b) It can be easily integrated with C, C++, COM, ActiveX, CORBA, and Java.
 - (c) Both of the above.
 - (d) None of the above.
3. Which of the following best describes the primary way in which a distributed denial-ofservice (D D o S) attack differs from a denial-ofservice (D o S) attack?
 - (a) The goal of the attack
 - (b) The number of computers being attacked
 - (c) The number of computers launching the attack
 - (d) The time period in which the attack occurs
4. In Python 3, what is ['a', 'Chemistry', 0, 1][-3]?
 - (a) a
 - (b) Chemistry
 - (c) 0
 - (d) 1
5. In Python 3, what is ['a', 'Chemistry', 0, 1][1]?
 - (a) a
 - (b) Chemistry
 - (c) 0
 - (d) 1

True / False (10 marks)

Answer True or False

6. True or False: The function $O(\log N)$ grows slower than $O(\log \log N)$ and $O(N)$.
7. True or False: `"abc"::-1` in Python 3 outputs `"cba"`, which is the string reversed.
8. True or False: The method `'ab'.sort()` can be used to sort the characters in the string `'ab'` directly.
9. True or False: In Python 3, the expression `'a' + 'ab'` evaluates to `'aab'`.
10. True or False: In Python 3, the expression `8 >> 1` evaluates to 3, which results from incorrectly calculating the right bitwise shift.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Explain the concept of floor division in Python 3, including its syntax and how it differs from regular division. Provide examples to illustrate your explanation.
12. Discuss why extensive testing of a large Java program may not guarantee the absence of bugs. What factors could lead to undetected issues in the software?

End of Paper